

Earth map + coord selection + heatmap overlay

- **Draw/bound regions:** Mapbox GL JS or Deck.gl + [mapbox-gl-draw](#) for the circle/polygon tool. For "fucking amazing," go **Deck.gl + Mapbox satellite basemap** (or CesiumJS if you want literal 3D globe — more wow, harder).
- **Heatmap layer:** GFW **4Wings API** returns gridded raster data for [AIS Apparent Fishing Effort](#), [AIS Vessel Presence](#), and [SAR Vessel Detections](#). You can request **HIGH** resolution (0.01°) and get it back as a heatmap-ready grid, grouped by [FLAG](#), [GEARTYPE](#), [MMSI](#), or [VESSEL_ID](#). Use the region codes from the References API (EEZ, MPA, RFMO, or your own shapefile) to scope the query — and that's the same polygon the user just drew on the map. This closes the loop cleanly.
- Gotcha: 4Wings date range is capped at **366 days per call**. Paginate if you want multi-year history.

Illegal-fishing calculation engine

This is the feature where you over-planned. GFW already has the scoring pipeline:

- **Insights API** — "designed to support risk-based decision-making ... for assessing risks of IUU fishing." It fuses AIS activity + public authorizations + AIS-off events into vessel-level "insights." This is the 80% solution for your risk classifier. Your agent wraps this with the *regional* overlay.
- **Events API** — gives you the discrete events your rules trigger on: [FISHING](#), [ENCOUNTER](#), [LOITERING](#), [PORT_VISIT](#), [GAP](#) (AIS-off). An AIS gap inside a no-take MPA is basically a one-line rule.
- **Vessels API** — flag state, IMO/MMSI/callsign, gear type, ownership (via [includes=OWNERSHIP](#)), and [AUTHORIZATIONS](#) (is this vessel licensed to fish here?). Unauthorized + fishing event = flag.

Regional law derivation (your "scrape the web for regional rules" piece): don't freestyle. Use FAO's databases — they've already collected this:

- **FAOLEX:** largest legislative database for fisheries/food/natural resources
- **FISHLEX:** specifically "Coastal State Requirements for Foreign Fishing" — literally a per-country table of what foreign vessels can/can't do in that country's waters
- **PORTLEX:** port state measures for IUU enforcement (hand-in-glove with your port denial feature)
- **ProtectedSeas Navigator:** 21,700 marine areas with pre-extracted regulations + a "Level of Fishing Protection" score — you may not even need to scrape raw law text, just query this

Pragmatic play for the demo: pre-scrape 3 regions (e.g., Galápagos Marine Reserve, Philippines EEZ, EU Mediterranean) into your agent's knowledge base, use FAOLEX fulltexts, have Gemini extract rules into JSON on the fly during the demo for "impressive live behavior."

Fine calculator

Inputs you need, and where each comes from:

- **Estimated catch volume:** fishing hours × gear-type catch-rate constants (literature values; FAO has them). Fishing hours directly from 4Wings. Gear type from Vessels API.
- **Species:** infer from region + gear type + season (fishing-zone species maps) or from catch-reporting data. Don't promise species-level ID from satellite — judges will ask and it isn't real.
- **Jurisdiction:** from EEZ polygon intersect (References API / Marine Regions) or flag state (Vessels API) depending on offense type.
- **Prior offenses:** Insights API + your own event log.
- **Fine schedule:** extracted from FAOLEX law text by Gemini. Output: PDF via [reportlab](#) or [weasyprint](#).

Direct vessel hail (VHF voice)

Real VHF from a hackathon = no. But you don't need real radio — you need a **convincing demo**. Two paths:

1. **ElevenLabs voiced cease-and-desist** playing in the UI as if it's being broadcast on Channel 16, with a waveform animation and "TRANSMITTING — CH 16 VHF — MMSI 412xxxxxx" overlay. Judges will 100% buy this as the demo.
2. **Fallback PDF cease-and-desist letter** — auto-generated, signed with agent metadata, timestamp, and the evidence bundle attached. This is what real authorities actually send anyway.

Do both. The voice is demo eye-candy, the PDF is the "real" artifact.

Port state denial (phone call agent)

- **Data:** NGA's **World Port Index** — free CSV, 3,818 ports, 106 fields including UN/LOCODE, coordinates, country. Doesn't include phone numbers for port authorities, but you don't need real ones — you need one demo number (a teammate's phone) that gets called live on stage.
- **Next-port prediction:** GFW Events API has [PORT_VISIT](#) events with history; simplest feasible approach is "vessel's most-frequented ports in last 90 days" → top candidates. You can dress this up with a trajectory-based prediction if time allows, but the simple version will land fine.

- **Call agent: Bland AI** is hackathon-optimal — literally ~10 lines of code to initiate an outbound call with a prompt + dynamic variables. Vapi if you want more plumbing control; Retell if latency matters (600ms vs 800ms, irrelevant for a demo).
- **Demo play:** "Next predicted port: Manta, Ecuador. Notifying port authority now." → phone rings on stage → Bland agent reads the scripted port-state denial notice with the vessel MMSI, offense, evidence URL. This is the single most demo-worthy moment in your whole pitch — make sure it's the climax.

Agent: historic + realtime overfishing chain-of-thought

All the signals are there:

- Historical: Vessels API + Events API (**FISHING**, **GAP**, **LOITERING**, **PORT_VISIT**) back to 2012
- "Realtime": ~72-hour data delay on events, ~5-day delay on SAR detections. You cannot say "live" — say **"near-real-time"**. Judges won't ding you for this; they'll ding you for claiming live when it isn't.

Agent: proof/evidence generation

Straightforward. For each flagged event, bundle:

- Satellite tile (Cloudinary host, link in the PDF)
- AIS track snippet (plot from Events API raw positions)
- Insights API risk score JSON
- Region boundary + regulation excerpt (from FAOLEX)
- SHA-256 hash of the bundle + timestamp → write the hash into the PDF footer as a pseudo-chain-of-custody. This is the "prosecutor-grade evidence" line that sells judges.

Note on SAR and your CV skills

Your instinct to train a YOLO on xView3-SAR is still valid for the narrative — **GFW's 4Wings SAR detections are already classified by a deep-learning model, so relying on them alone makes your CV work invisible to judges.** Recommended split:

- **Live pipeline:** consume GFW's SAR detections via 4Wings (reliable, no training risk)
- **CV showcase:** train YOLOv8-s on xView3-SAR on Kaggle P100, run inference live on a pre-downloaded Sentinel-1 tile during the demo to *visibly* detect dark vessels, then cross-reference with GFW AIS to confirm the dark-vessel-in-MPA narrative. This is your CV flex and makes the detection feel tangible.

One risk to surface now

Rate limits. GFW's 4Wings report endpoint throws 429 on concurrent requests per user and 524 (gateway timeout) if a report takes >100s. Your demo cannot hang. Pre-cache the demo scenario's API responses into local JSON and have your agents read from those during the live demo, with a "run live query" toggle for Q&A. Every winning hackathon demo I've seen does this.